Canadian Financials Penalized Regression

This report focuses on a Canadian financials cross-sectional ranking problem. The model uses a long monthly history, BoC and Statistics Canada data, Canadian market context, and stock-specific momentum and volatility features to rank the next month's names.

Data horizon: 2000-01-01 onward for stock and market prices, with macro inputs lagged one month for causality. The model evaluates 21 candidate configurations across multiple universes and target definitions before selecting the best one by validation rank IC and long-short quality.

Metric	Value
Best model	lasso / full
Best universe	banks_relative_sector
Target	relative_sector
Mean rank IC	0.0995
Mean hit rate	22.76%
Model long-short	0.3137
Momentum long-short	-0.0044
Forecast rows	1,044
Date range	2011-12-31 to 2026-05-31
Selected features	131

How penalized regression works

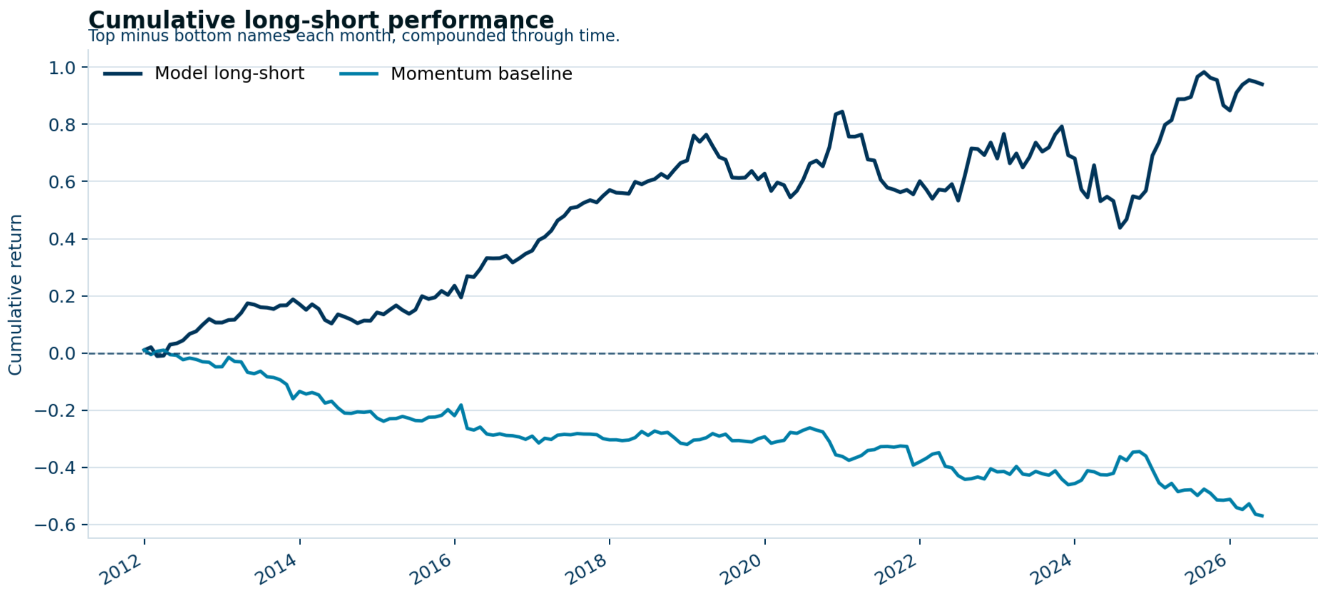
Penalized regression is ordinary regression with an added cost for large coefficients. The penalty shrinks weak or redundant variables toward zero, and in the lasso case some coefficients are pushed all the way to zero so they drop out of the model entirely.

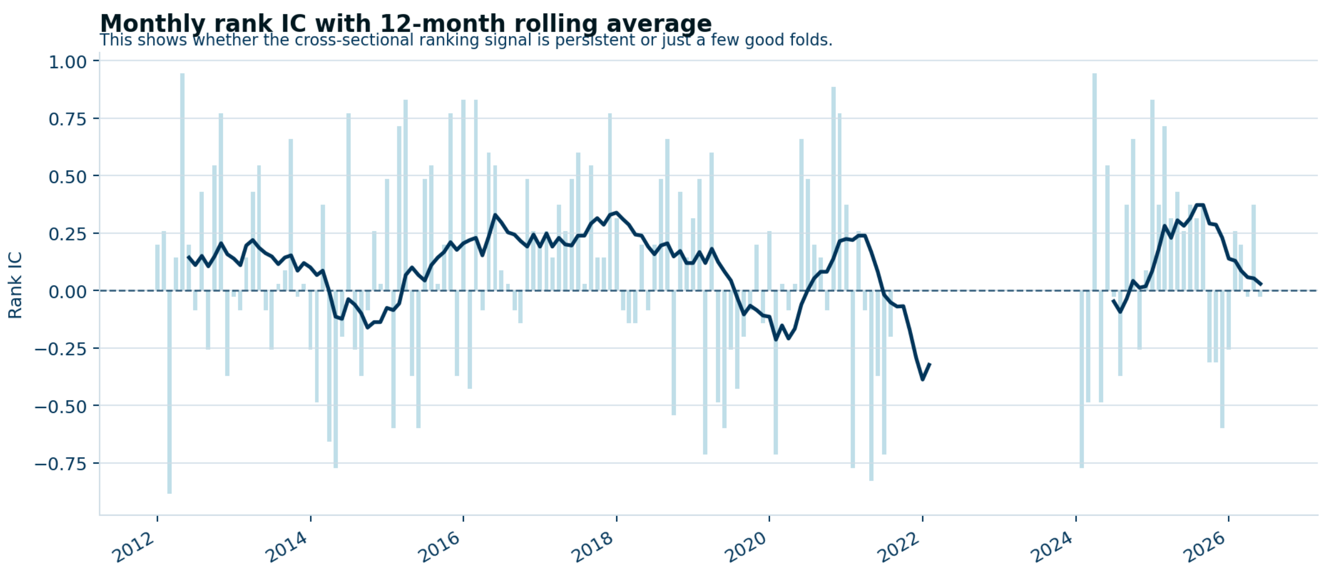
That matters in finance because signals are usually correlated, noisy, and fragile. A penalty makes the model favor a smaller set of variables that still explain the ranking problem out of sample. The coefficients keep their usual meaning as direction and relative strength, but now they are chosen with more discipline than plain OLS.

In this Canadian financials run, the model uses rate-sensitive macro data, stock momentum, and sector-relative context to rank next-month performance across Canadian banks. The point is not a single forecast number; it is a cleaner ranking, a readable long-short spread, and a shortlist of variables that survived the penalty.

Model and strategy evidence

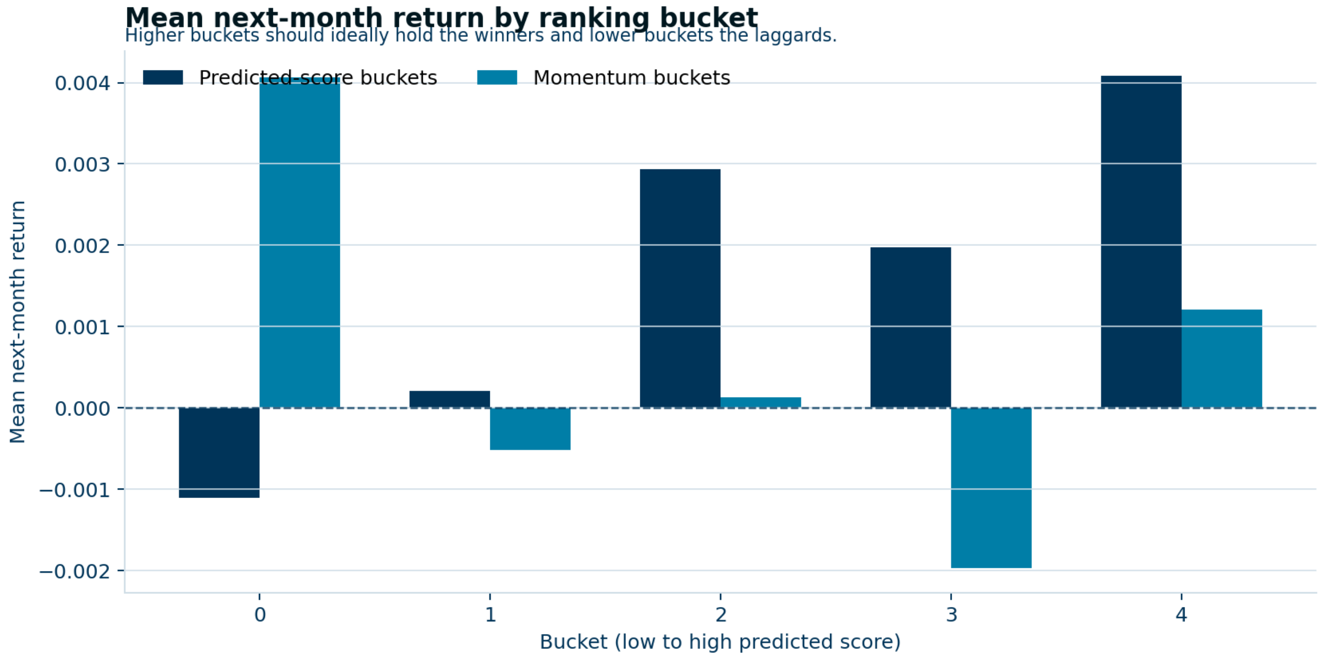
The first chart shows whether the selected model actually beats a simple momentum baseline on the long-short portfolio. The second chart checks whether the rank signal is steady month to month or just a few lucky folds.





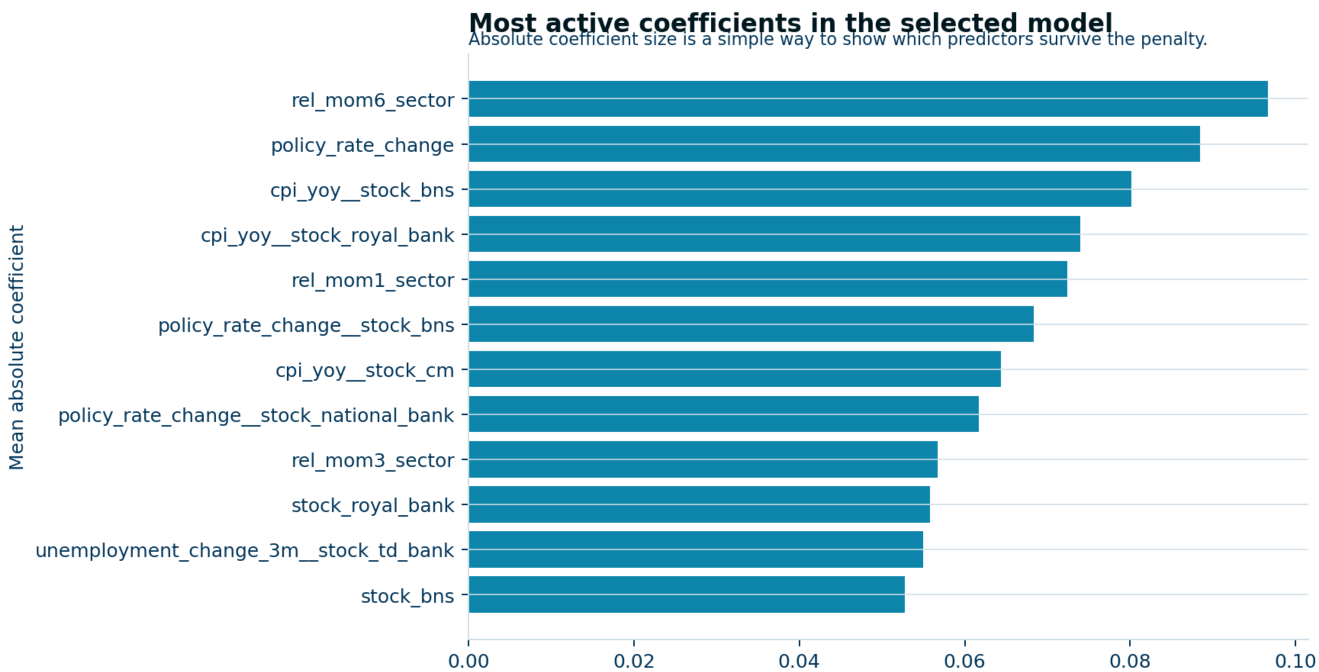
Where the signal shows up

The bucket chart turns the ranking problem into an intuitive test: the highest predicted buckets should hold the better forward returns, and the momentum baseline is plotted beside it for reference. If the model is useful, the top buckets should separate from the bottom buckets more cleanly than the baseline.



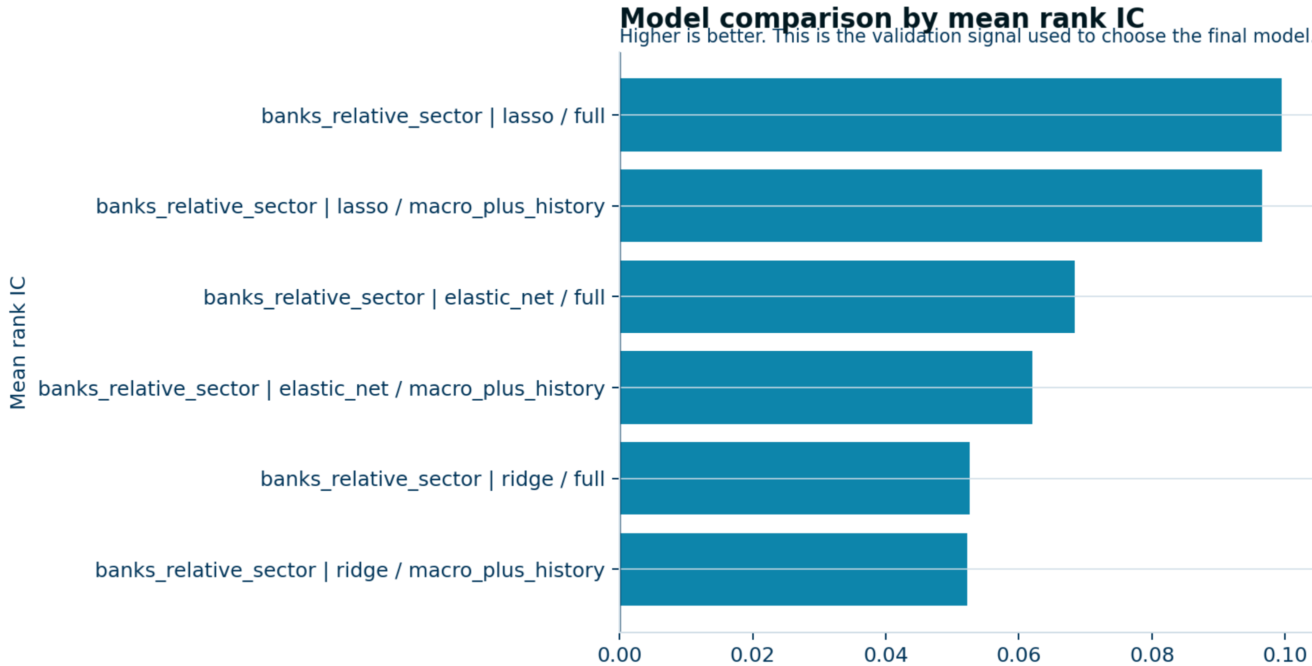
Which variables survived the penalty

The final chart shows the predictors that stayed active most often across folds. A useful penalized model should keep the story small: a few rate, momentum, relative-strength, and sector-context variables should dominate if the signal is real.



Model comparison

The table below shows every candidate configuration. This is the check that keeps us from over-selling one run: the selected model has to win against simpler penalties and narrower feature sets.



universe	target	model	features	rank_ic	hit_rate	ls_mean	mom_ls_mea
banks_relative_secto	relative_sector	lasso	full	0.0995	22.76%	0.3137	-0.0044
banks_relative_secto	relative_sector	lasso	macro_plus_history	0.0966	28.74%	0.4708	-0.0044
banks_relative_secto	relative_sector	elastic_net	full	0.0684	23.45%	0.2598	-0.0044
banks_relative_secto	relative_sector	elastic_net	macro_plus_history	0.0621	28.16%	0.4183	-0.0044
banks_relative_secto	relative_sector	ridge	full	0.0525	23.56%	0.2289	-0.0044

Fold metrics

These are the walk-forward diagnostics for the selected configuration. The table keeps the model honest by showing how the signal behaves fold by fold rather than just in aggregate.

fold	test_start	test_end	alpha	rank_ic	hit_rate	ls_mean	mom_ls_mean
1	2011-12-31	2014-04-30	0.10000	0.0522	10.34%	0.0363	-0.0064
2	2014-05-31	2016-09-30	0.02848	0.1527	24.14%	0.5725	-0.0048
3	2016-10-31	2019-02-28	0.00811	0.2138	31.03%	0.6238	-0.0005
4	2019-03-31	2021-07-31	0.00811	-0.0404	13.79%	-0.3018	-0.0009
5	2021-08-31	2023-12-31	0.10000	nan	nan%	nan	-0.0068
6	2024-01-31	2026-05-31	0.00231	0.1192	34.48%	0.6380	-0.0071

Top coefficients

The selected features table below shows the coefficient list from the chosen run. This is the transparency layer behind the feature-importance chart.

feature	mean_coefficient	mean_abs_coefficient	selection_count
rel_mom6_sector	-0.09670	0.09670	4
policy_rate_change	-0.08849	0.08849	2
cpi_yoy__stock_bns	-0.08014	0.08014	1
cpi_yoy__stock_royal_bank	0.07395	0.07395	1
rel_mom1_sector	-0.07242	0.07242	5
policy_rate_change__stock_bns	0.06836	0.06836	3
cpi_yoy__stock_cm	-0.06437	0.06437	3
policy_rate_change__stock_national_bank	0.05650	0.06173	5
rel_mom3_sector	0.05674	0.05674	3
stock_royal_bank	-0.05577	0.05577	3
unemployment_change_3m__stock_td_bank	-0.05501	0.05501	4
stock_bns	-0.05279	0.05279	4
term_proxy	-0.04813	0.04813	1
financials_etf_vol12	-0.04763	0.04763	1
term_proxy__stock_td_bank	0.04699	0.04699	1